

Indicators of Heart Disease

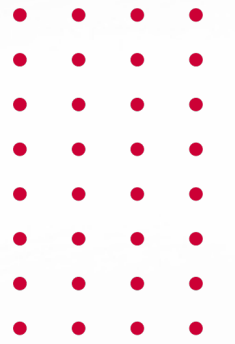
Binary Classification for Heart Disease Prediction

Presented by Chloe Barker and Jack Pehlke





Overview



MOTIVATION

Early prediction of heart disease can guide timely interventions, reduce healthcare costs, and improve patient outcomes.



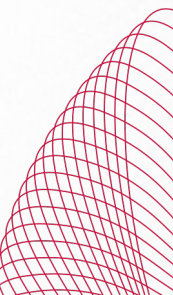
OBJECTIVE 1

Build an interpretable multiple logistic regression model using EDA-guided predictors, interpreting coefficients, confidence intervals, and practical vs. statistical significance.



OBJECTIVE 2

Compare baseline logistic regression to additional classification models (complex logistic, LDA, QDA, nonparametric) and recommend the best-performing model based on key performance metrics.



Data Overview



 **Dataset:** Indicators of Heart Disease (2020) from Kaggle

 **Source:** 2020 annual CDC BRFSS survey data (300k+ adults) on their health status and behaviors

- 14 Categorical Predictors, 4 Numeric Predictors

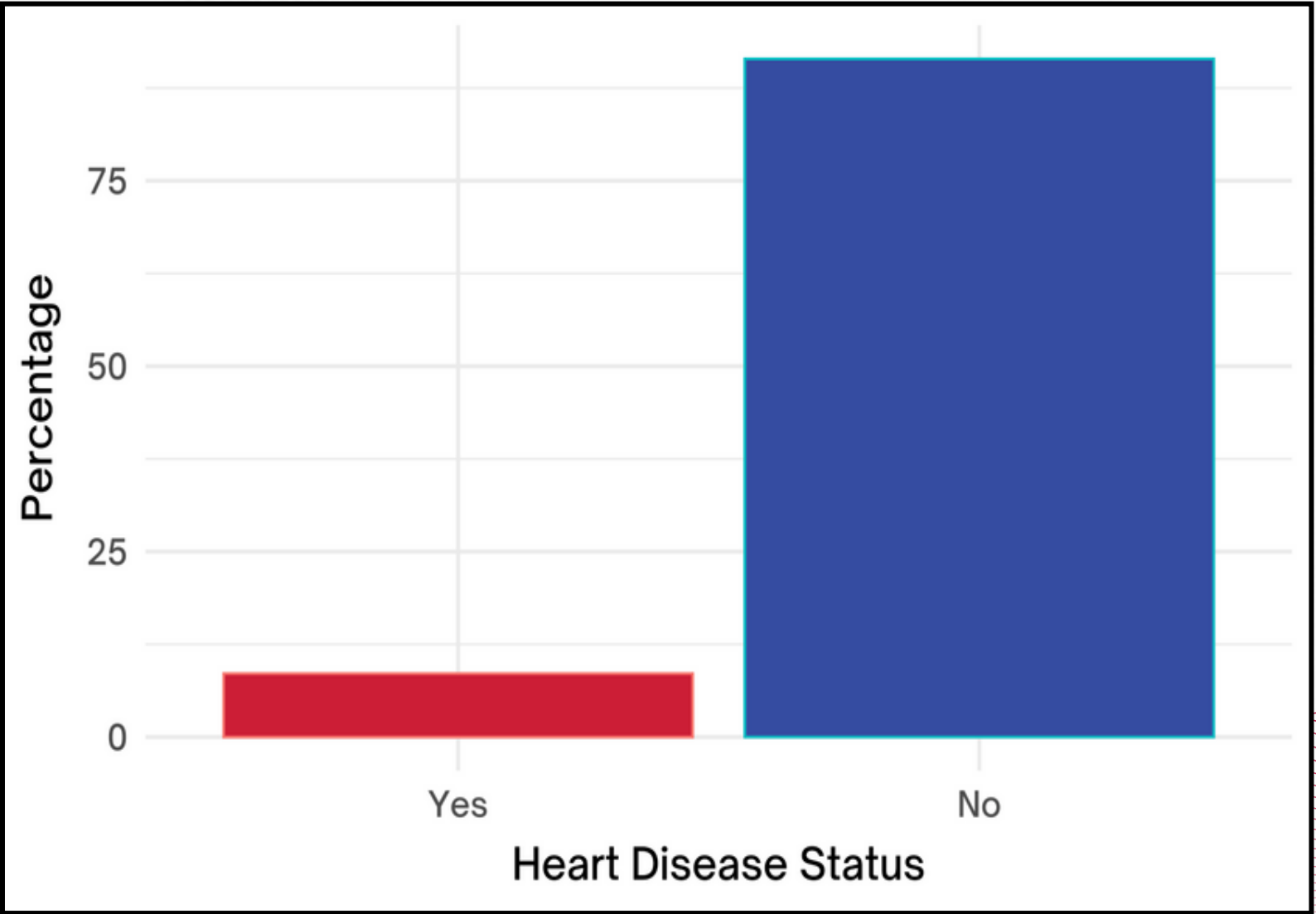
 **Target:** Heart Disease

- "Yes" - respondent had heart disease
- "No" - respondent did not have heart disease

Heart Disease Sample Size (%)

Full Dataset

Heart Disease	Sample Size	%
Yes	27,373	8.56
No	292,422	91.4



Training Dataset

Heart Disease	Sample Size	%
Yes	21,899	8.56
No	233,938	91.4



Categorical Variables



Binary

Variable Name	Detailed Description
Heart Disease	Target trait.
Smoking	Do you smoke? (No, Yes)
Alcohol Drinking	Do you drink? (No, Yes)
Stroke	Have you been told you had a stroke? (No, Yes)
Difficulty Walking	Difficulty climbing stairs? (No, Yes)
Sex	Gender of a person. (Female, Male)
Physical Activity	Reported doing physical activity or exercise in the past 30 days outside of regular job. (No, Yes)
Asthma	Presence of asthma. (No, Yes)
Kidney Disease	Presence of kidney disease. (No, Yes)
Skin Cancer	Presence of skin cancer. (No, Yes)

3+ Levels

Variable Name	Detailed Description	Response Measure
Diabetic	Presence of diabetes or related condition.	No; No, borderline diabetes; Yes; Yes (during pregnancy)
General Health	Self-reported overall well-being.	Excellent, Fair, Good, Poor, Very good
Race	Race/ethnicity of respondent.	American Indian/Alaskan Native, Asian, Black, Hispanic, Other, White
Age Category	Age category of the subjects.	18-24, 25-29, ..., 80 or older

Continuous Variables

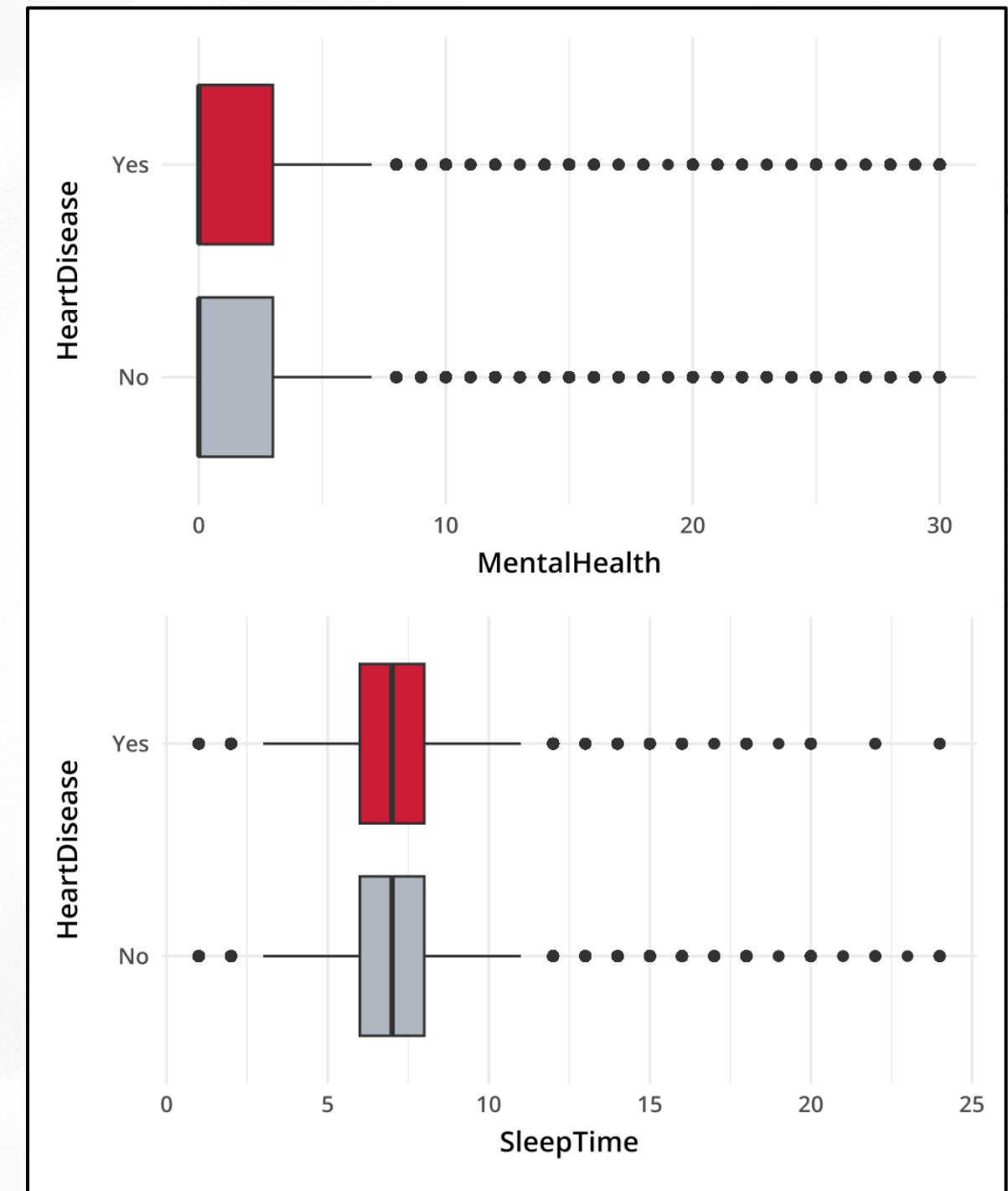
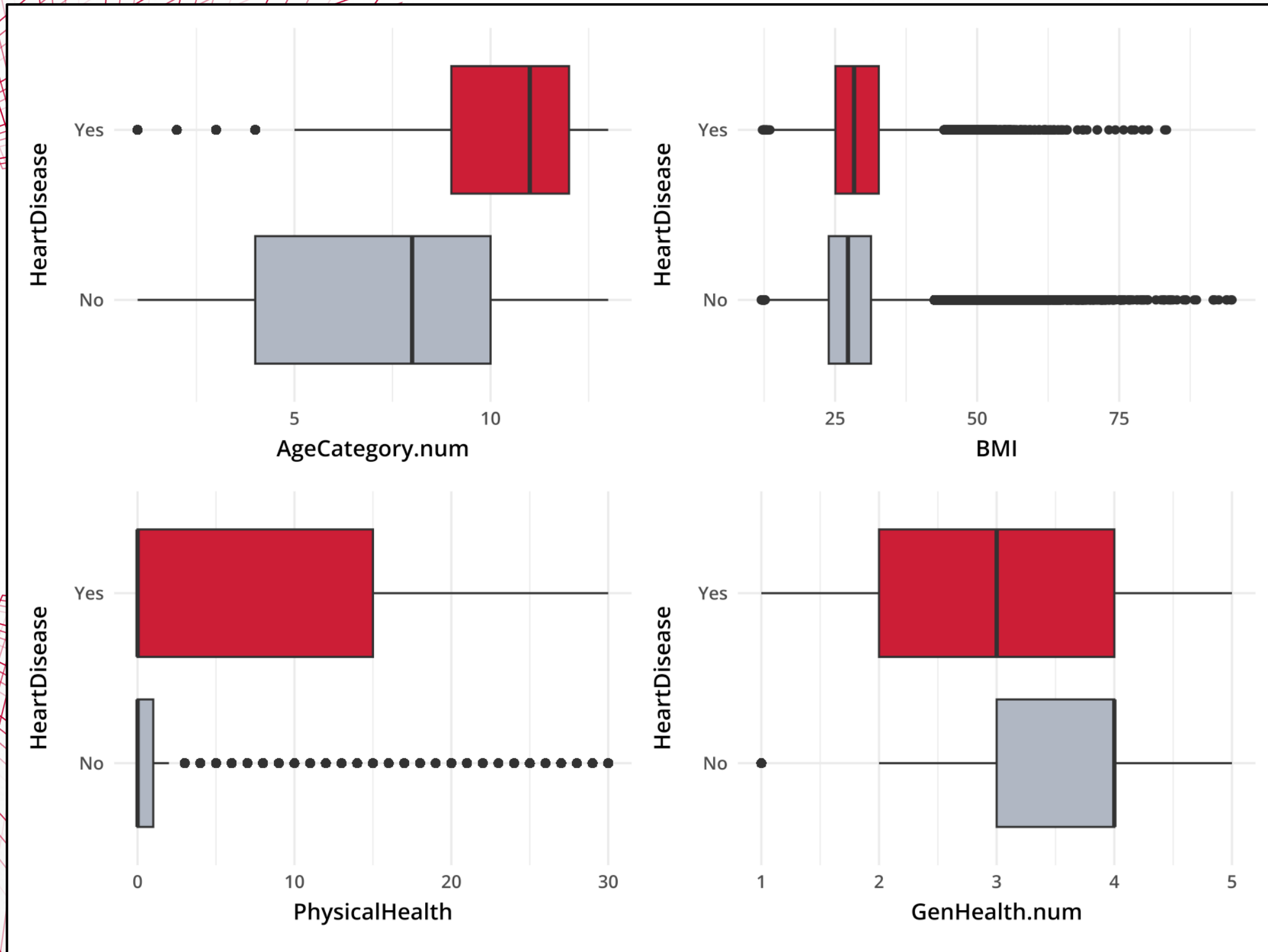


Variable Name	Detailed Description
BMI	Value assessing correspondence between height and mass to indicate underweight, normal, or overweight. (body mass index)
PhysicalHealth	Number of days in the past month with poor physical health. (0-30 days)
MentalHealth	Number of days in the past month with poor mental health. (0-30 days)
SleepTime	Average number of hours of sleep per night.
GenHealth.num	Categories: Poor (1) – Excellent (5)
AgeCategory.num	Age category made numeric (1-13)

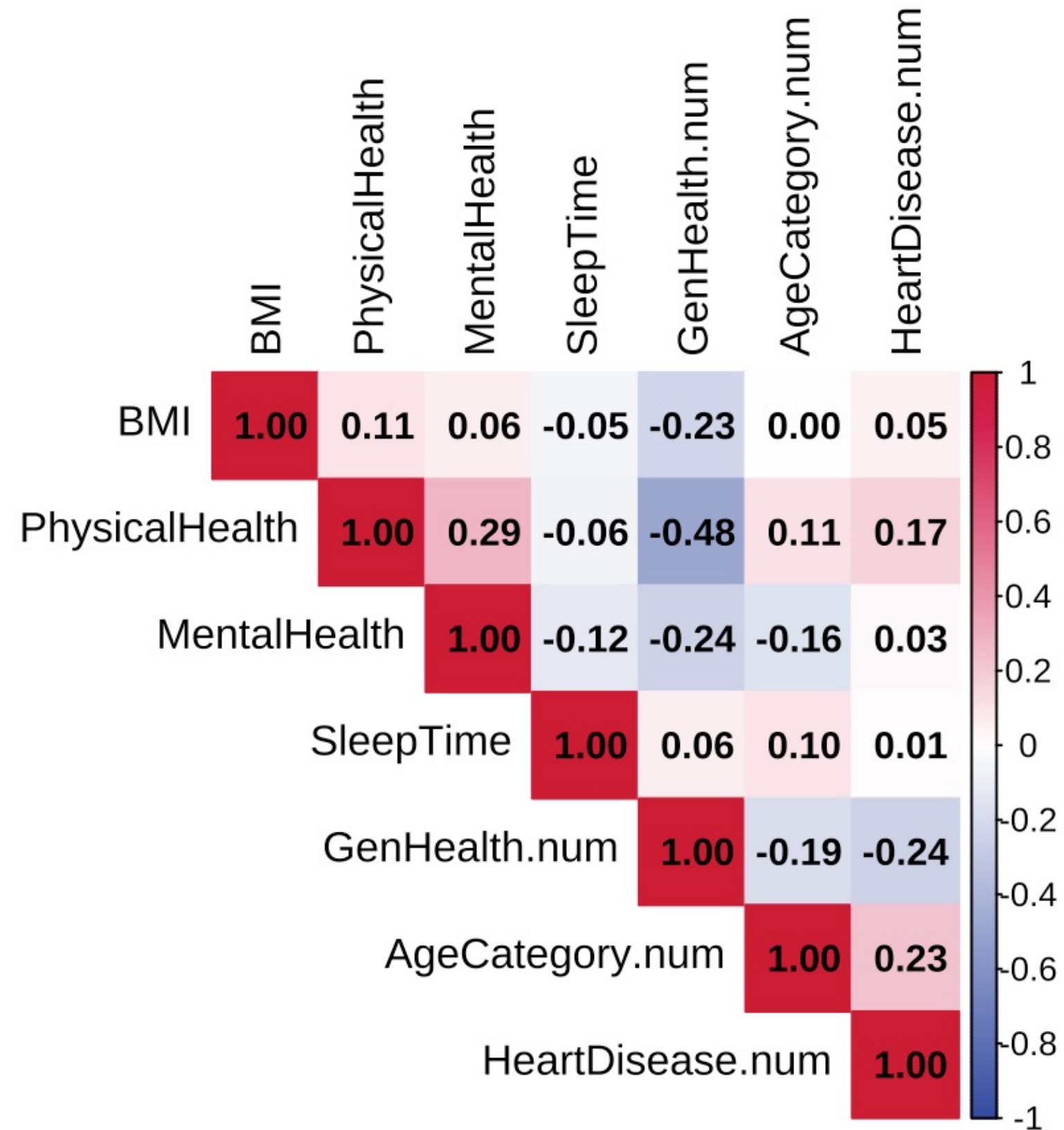
Statistic	BMI		Physical Health		Mental Health		Sleep Time	
Min.	12.21	12.02	0	0	0	0	1	1
1st Qu.	25.06	23.88	0	0	0	0	6	6
Median	28.34	27.26	0	0	0	0	7	7
Mean	29.42	28.22	7.833	2.951	4.64	3.823	7.128	7.092
3rd Qu.	32.69	31.32	15	1	4	3	8	8
Max.	83.33	94.85	30	30	30	30	24	24

Statistic	GenHealth.num		AgeCategory.num	
Min.	1	1	1	1
1st Qu.	2	3	9	4
Median	3	4	11	8
Mean	2.763	3.673	10.23	7.26
3rd Qu.	4	4	12	10
Max.	5	5	13	13

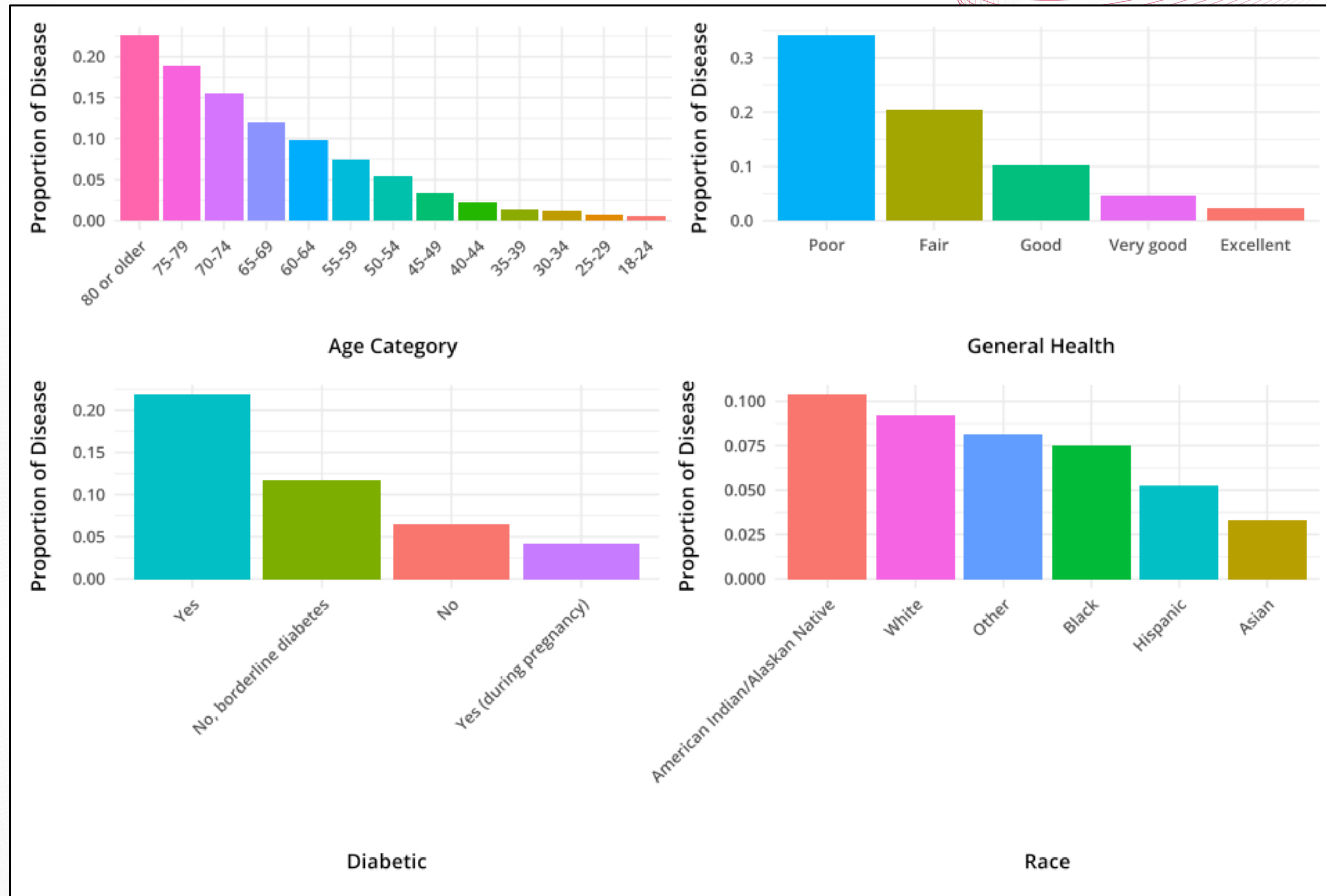
Exploratory Analysis



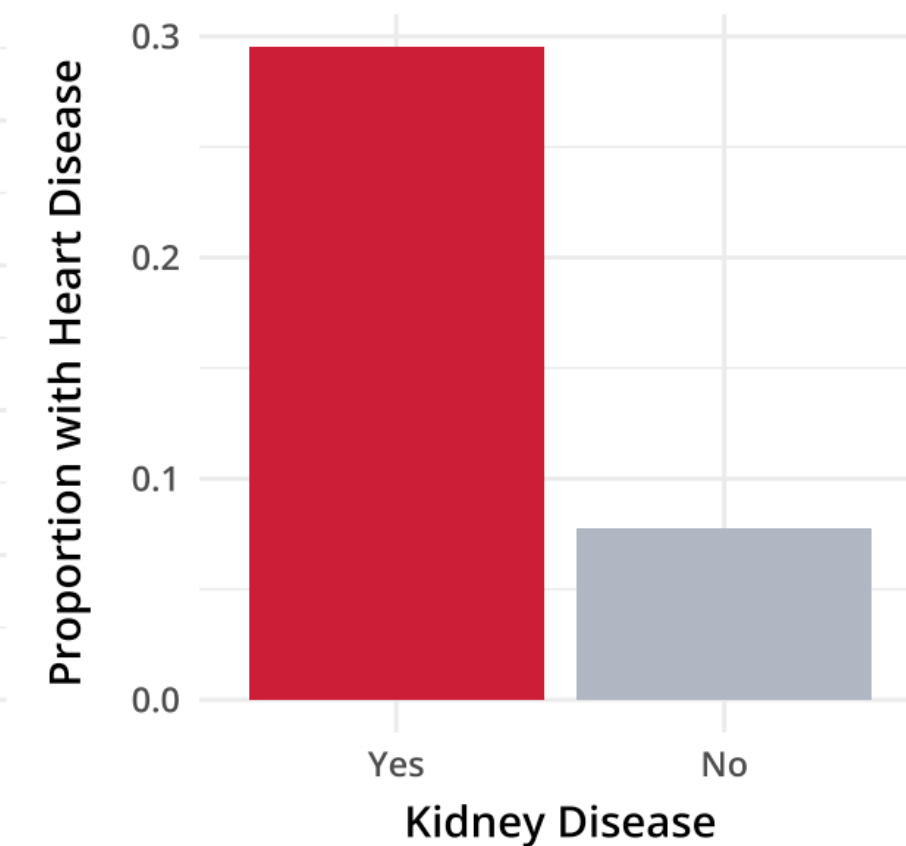
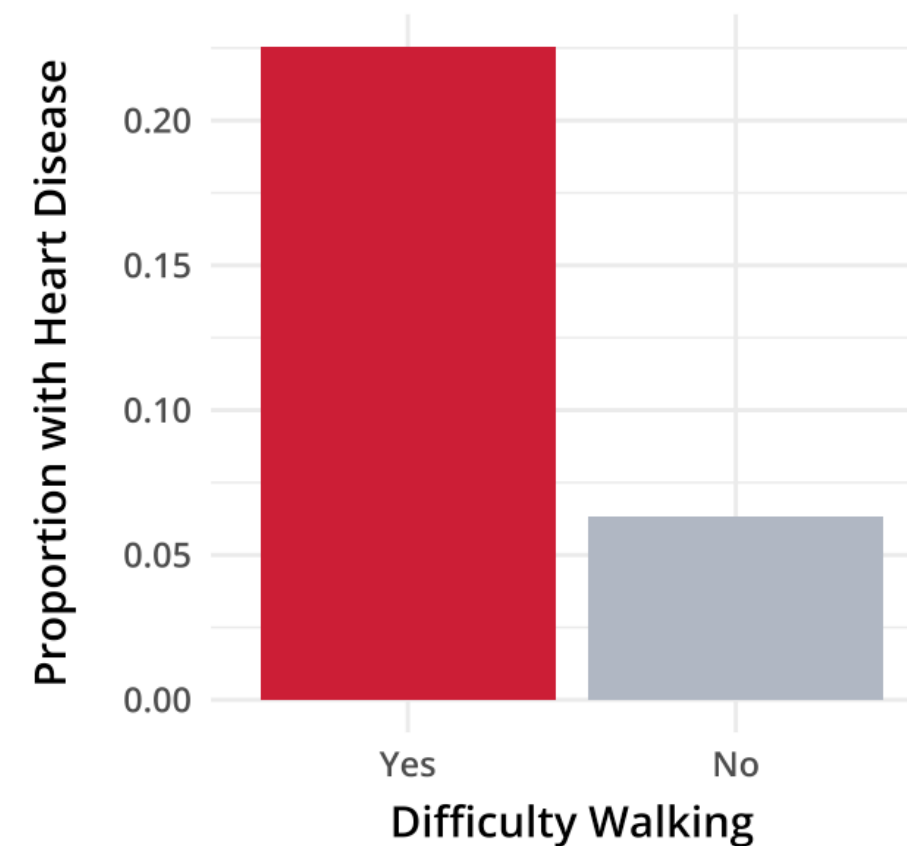
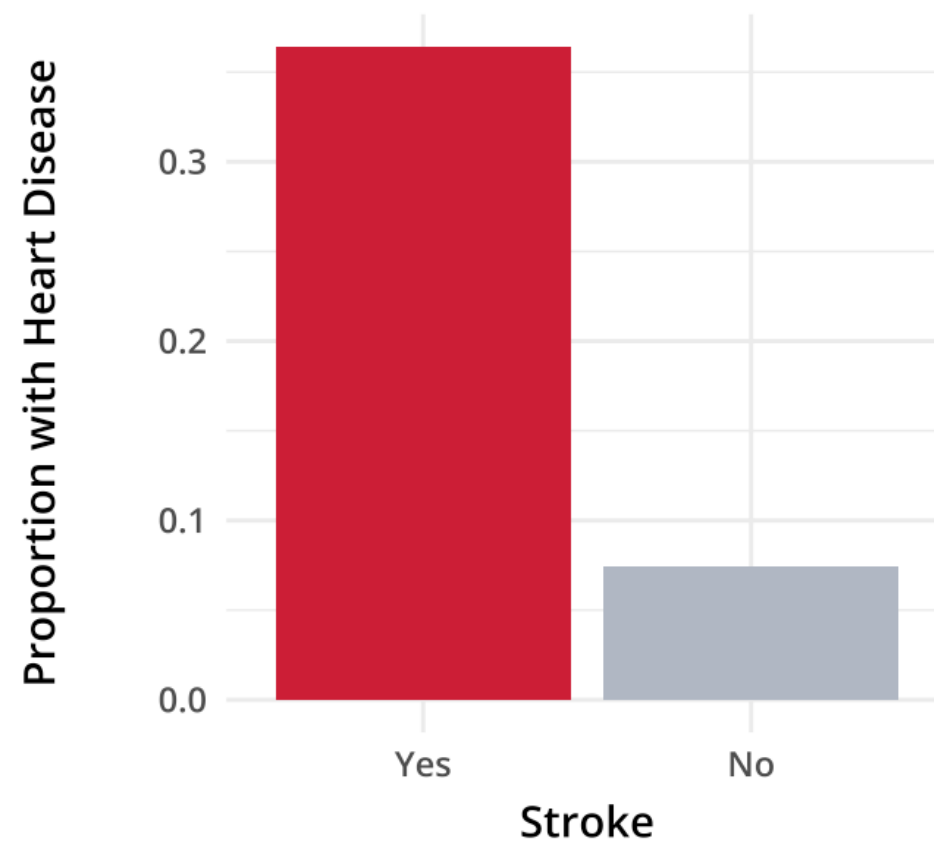
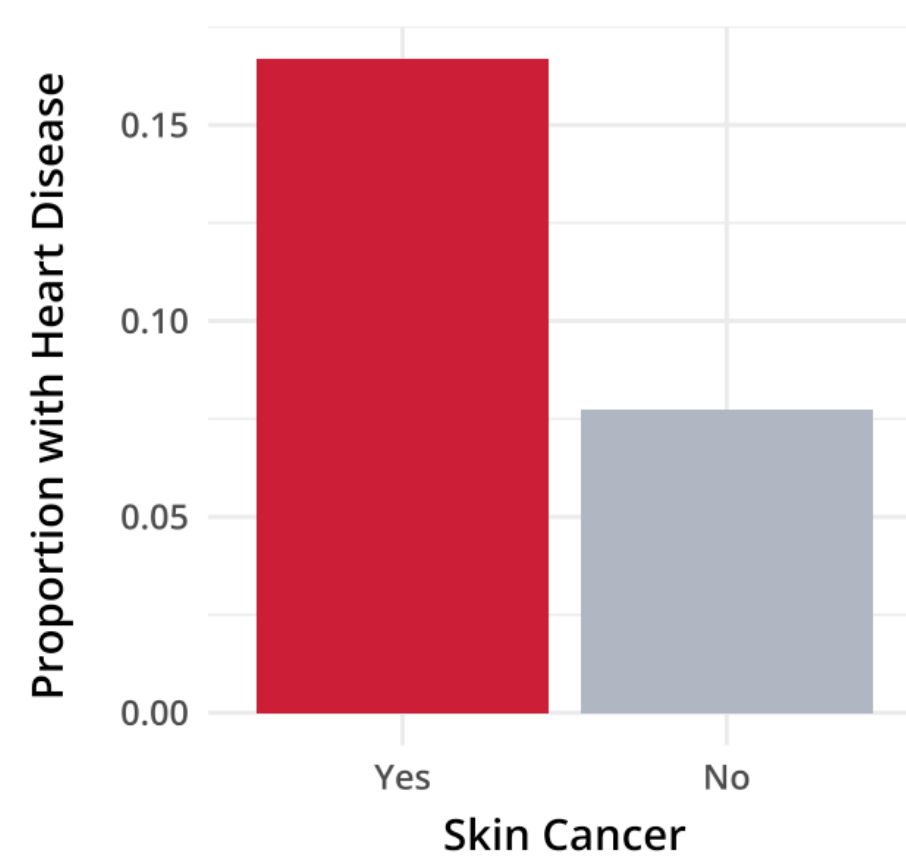
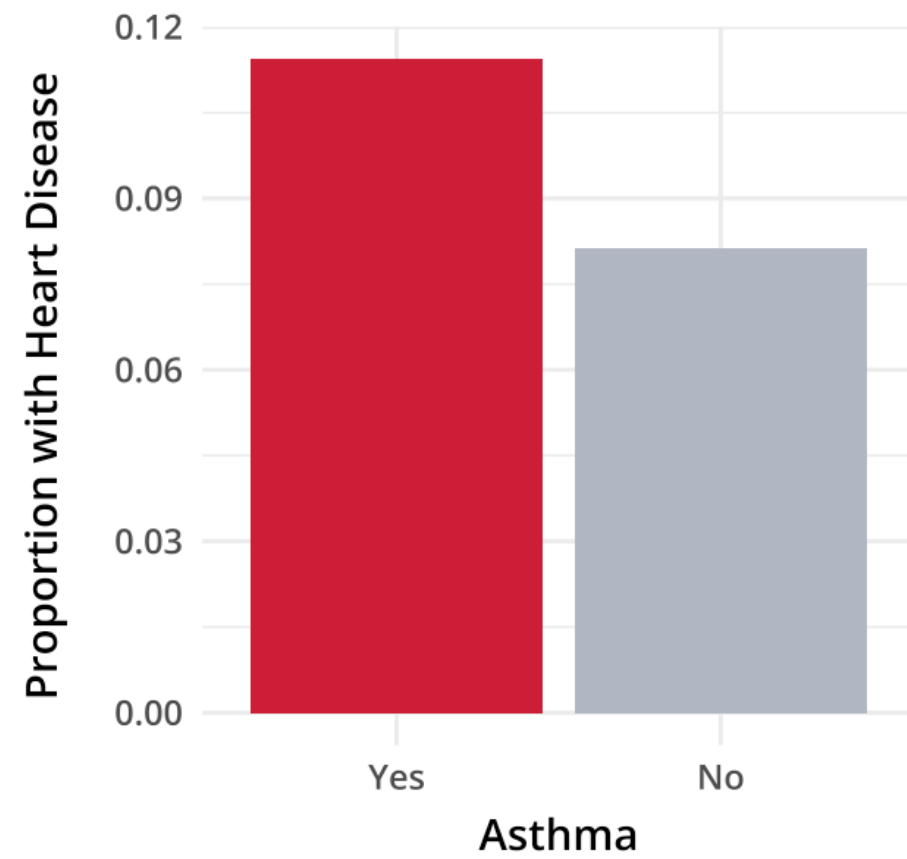
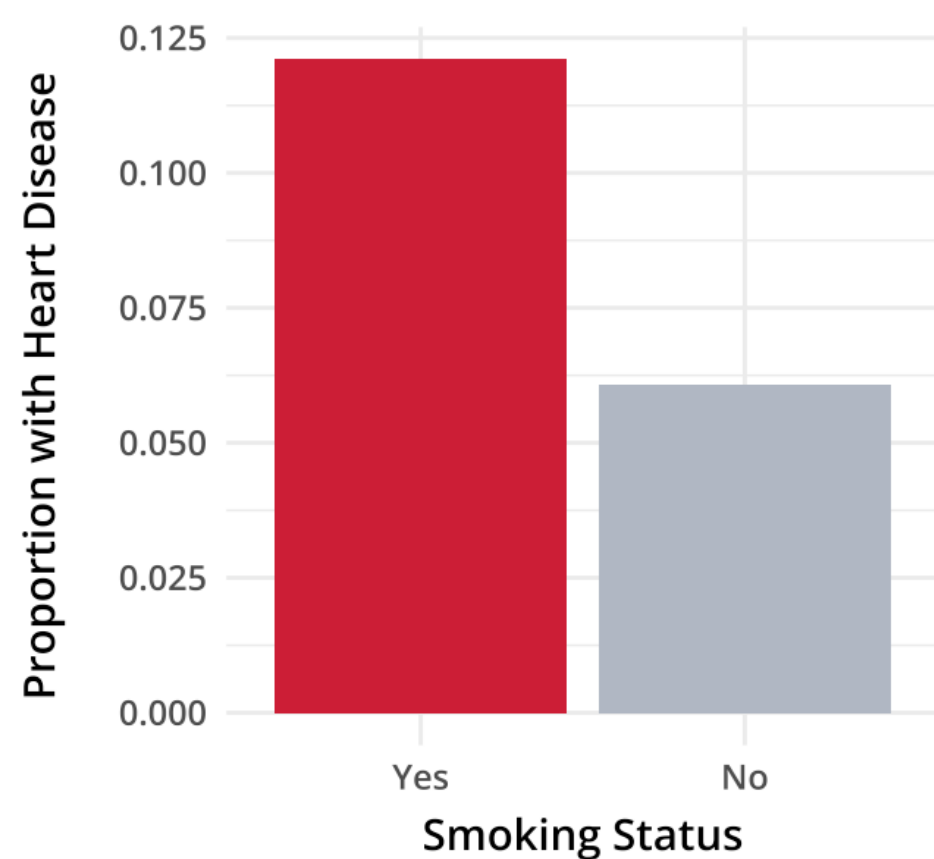
EDA: Continuous Predictors



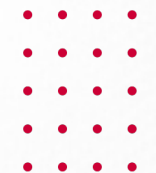
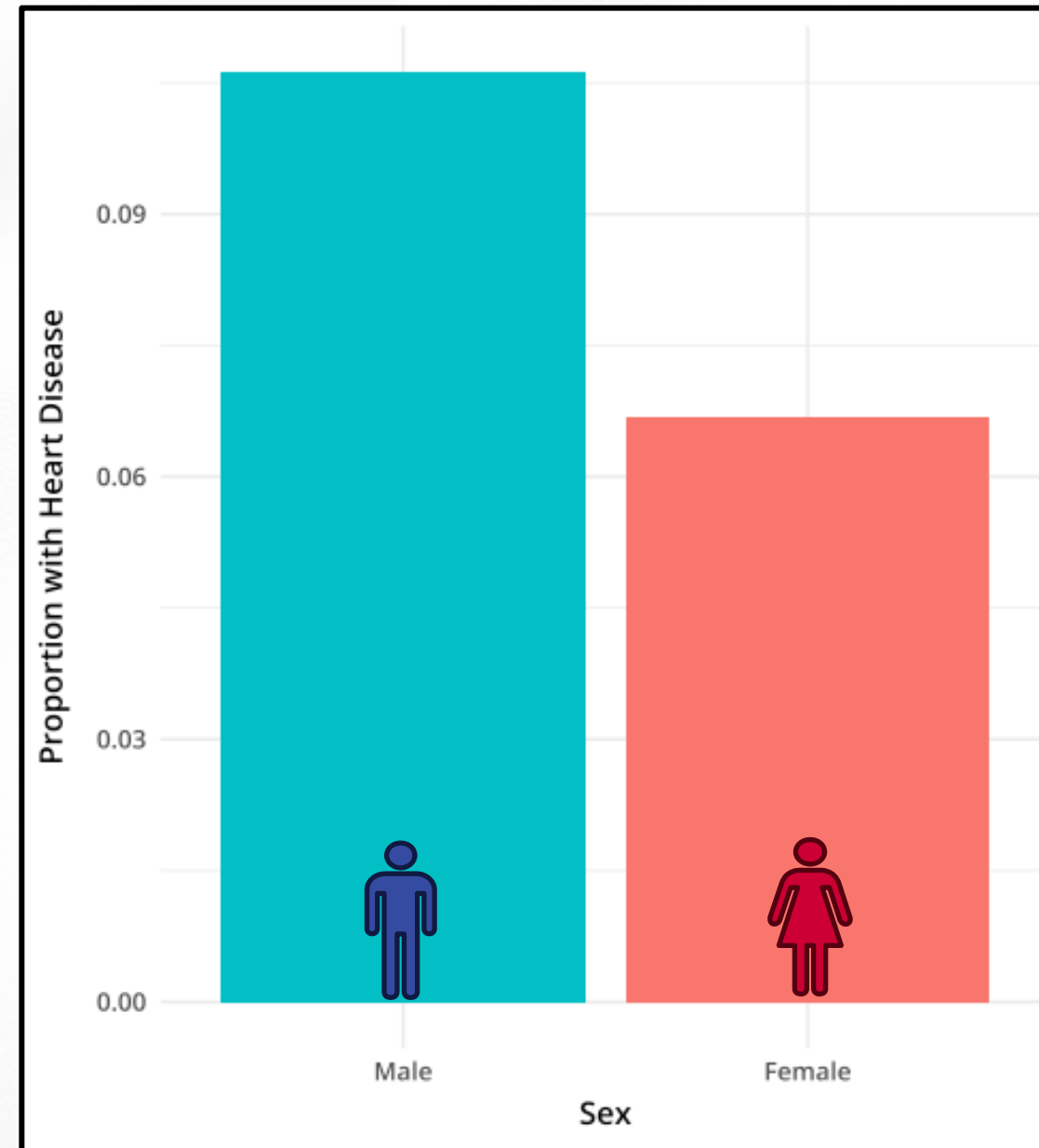
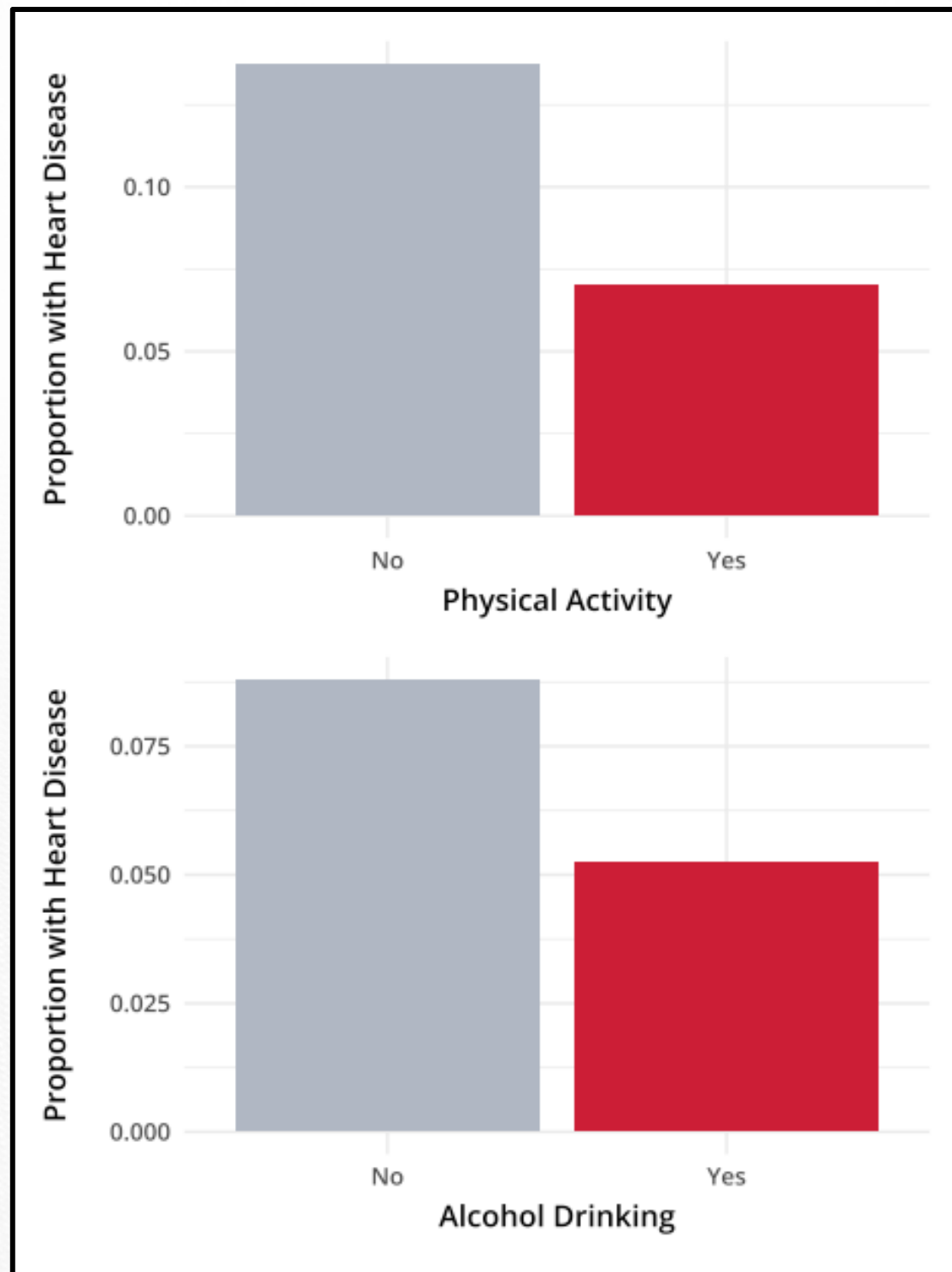
EDA: Categorical Predictors



Categorical Predictors Continued



Categorical Predictors Continued



Additive Logistic Regression

- **Objective 1:** Build a logistic regression model for interpretation:

Numeric Predictors

- PhysicalHealth
- BMI
- MentalHealth
- SleepTime

Categorical Predictors

- AgeCategory
- GenHealth
- Sex
- Race
- Smoking
- AlcoholDrinking
- Stroke
- DiffWalking
- Diabetic
- Asthma
- KidneyDisease
- SkinCancer

Removed: PhysicalActivity (not statistically significant)

AIC:

→ Full Model: 115991

→ Statistical Significance:

- All variables included except Physical Activity were found to be statistically significant.

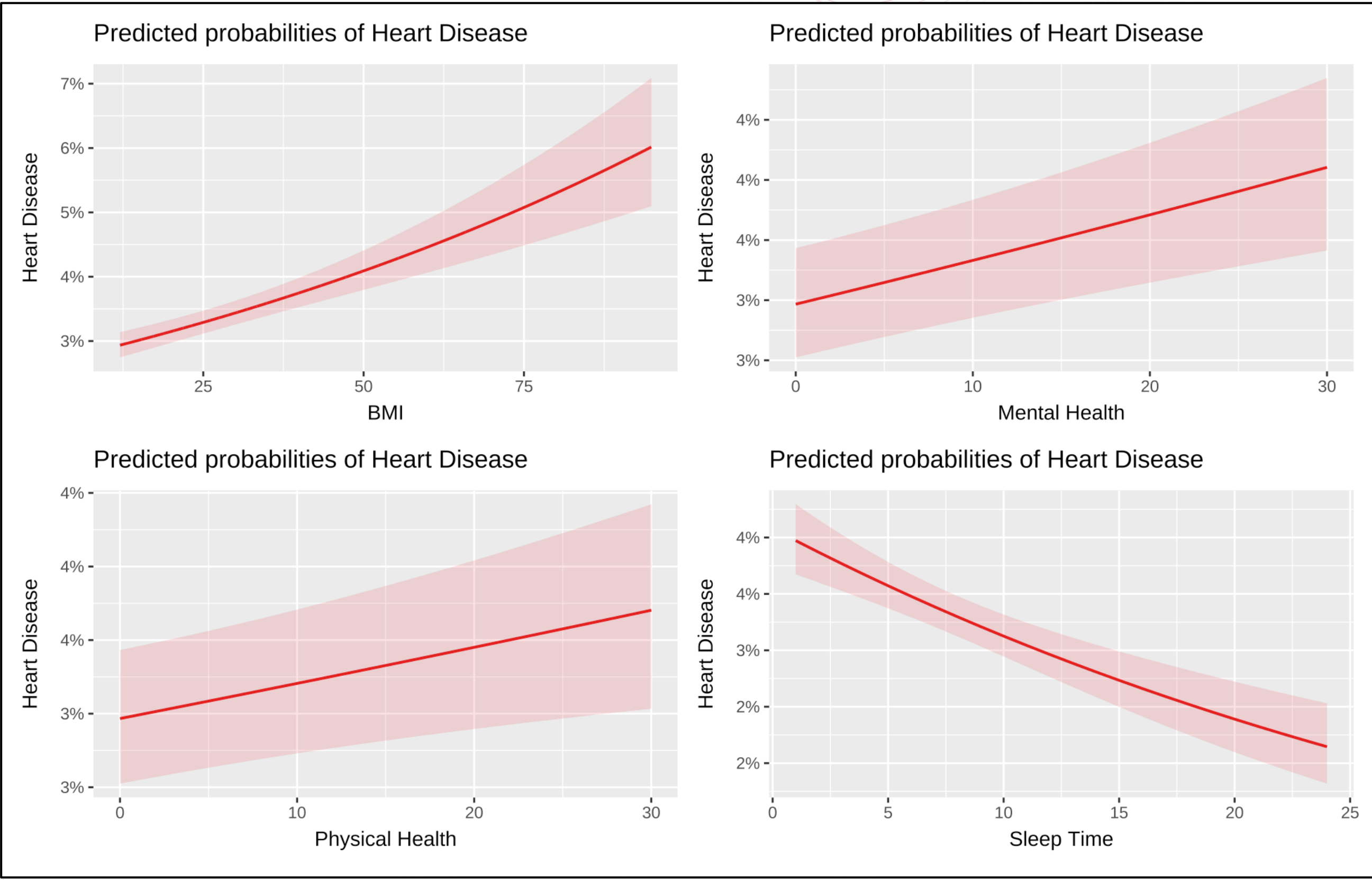
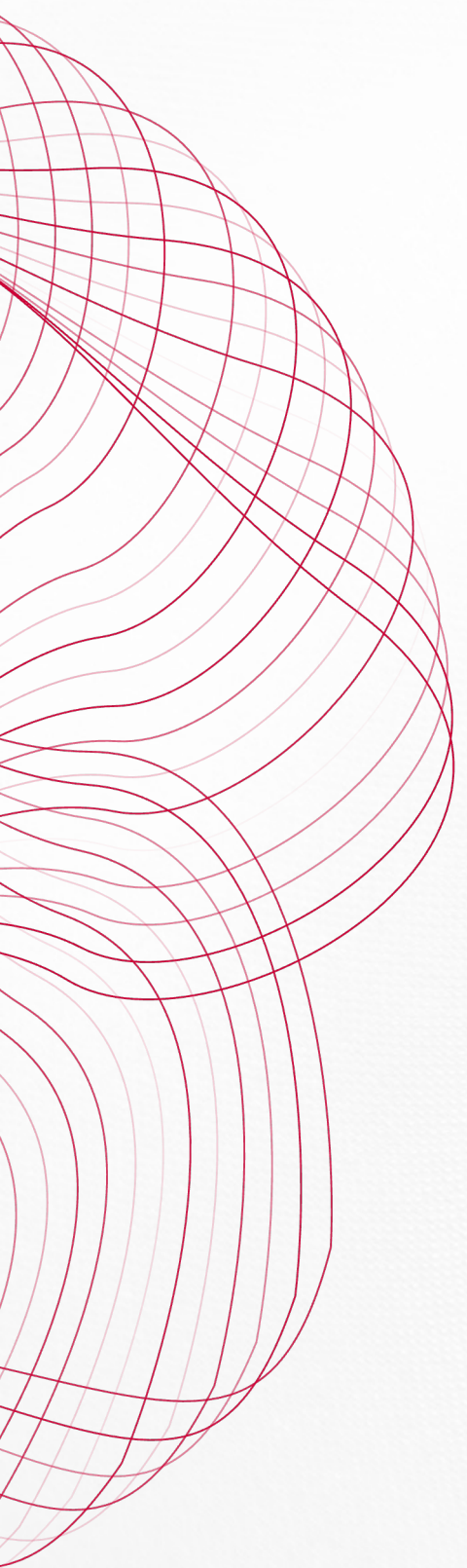
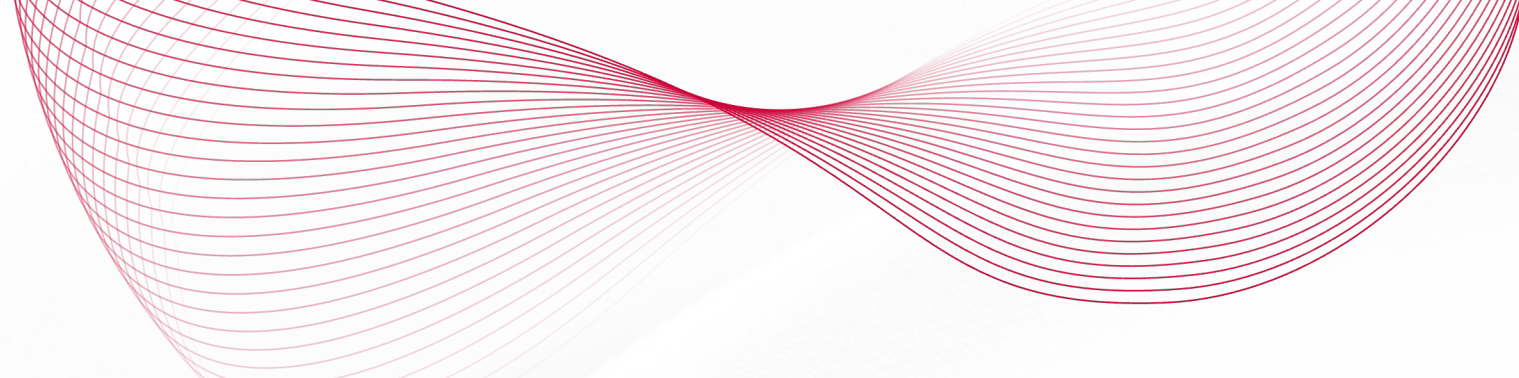
→ VIF values were all less than 2 ($VIF < 2$)

→ *Reduced model without Physical Activity* : 115989.1

→ **Practical Significance:** Because physical activity likely is known to effect heart disease and the difference in AIC is minimal, we will live it in the model so it can be accounted for. All variables were found to be practically significant.



Effect Plots



Interpretation of our Model



Numeric Predictor: BMI

Using the logistic regression summary table, there is sufficient evidence to conclude that the regression coefficient for BMI is different from 0 (p-value<0.001).

For a 1 unit increase in **BMI**, the odds of having Heart Disease increase by a factor of $e^{0.00912}=1.009$. (95% CI: (1.007, 1.012)).

Interpretation of our Model

Categorical Predictor: General Health

Upon examination of the regression coefficient tests, there is sufficient evidence to conclude that each coefficient for each General Health level is different from 0.

The odds of a person having Heart Disease with Very Good general health is 1.566 times higher than the odds for a person with Excellent general health. (95% CI: (1.466, 1.674))

The odds of a person having Heart Disease with Good general health is 2.789 times higher than the odds for a person with Excellent general health. (95% CI: (2.615, 2.975))

The odds of a person having Heart Disease with Fair general health is 4.504 times higher than the odds for a person with Excellent general health. (95% CI: (4.193, 4.839))

The odds of a person having Heart Disease with Poor general health is 6.568 times higher than the odds for a person with Excellent general health. (95% CI: (6.006, 7.183))

Objective 2 Models Used

Build models that prioritize prediction performance.



Complex Logistic Regression

Parametric – Model the mean of the response (expected value), given X predictors



KNN

Non-Parametric – Classify observations based on k number of nearest neighbors



LDA

Parametric

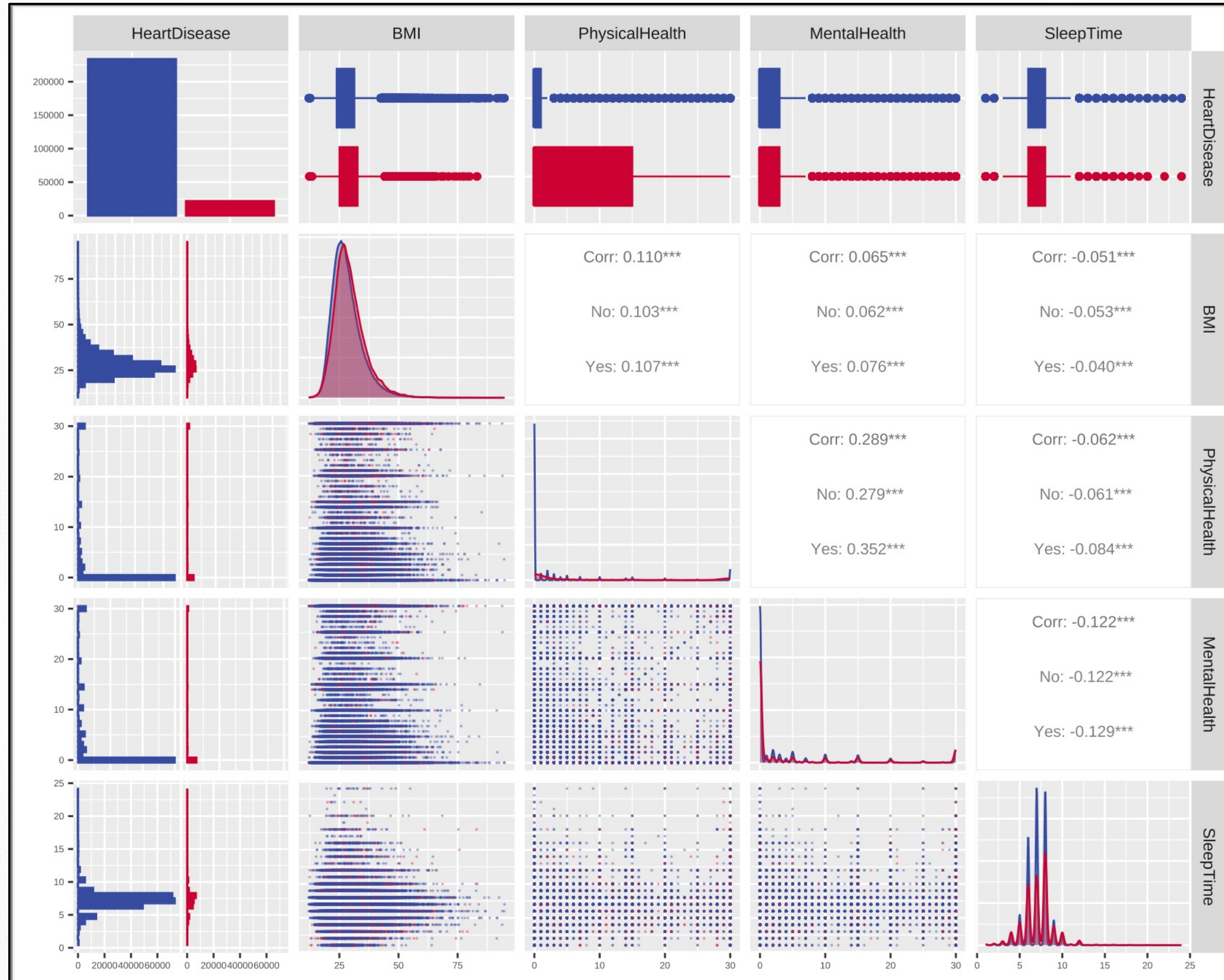


QDA

Parametric

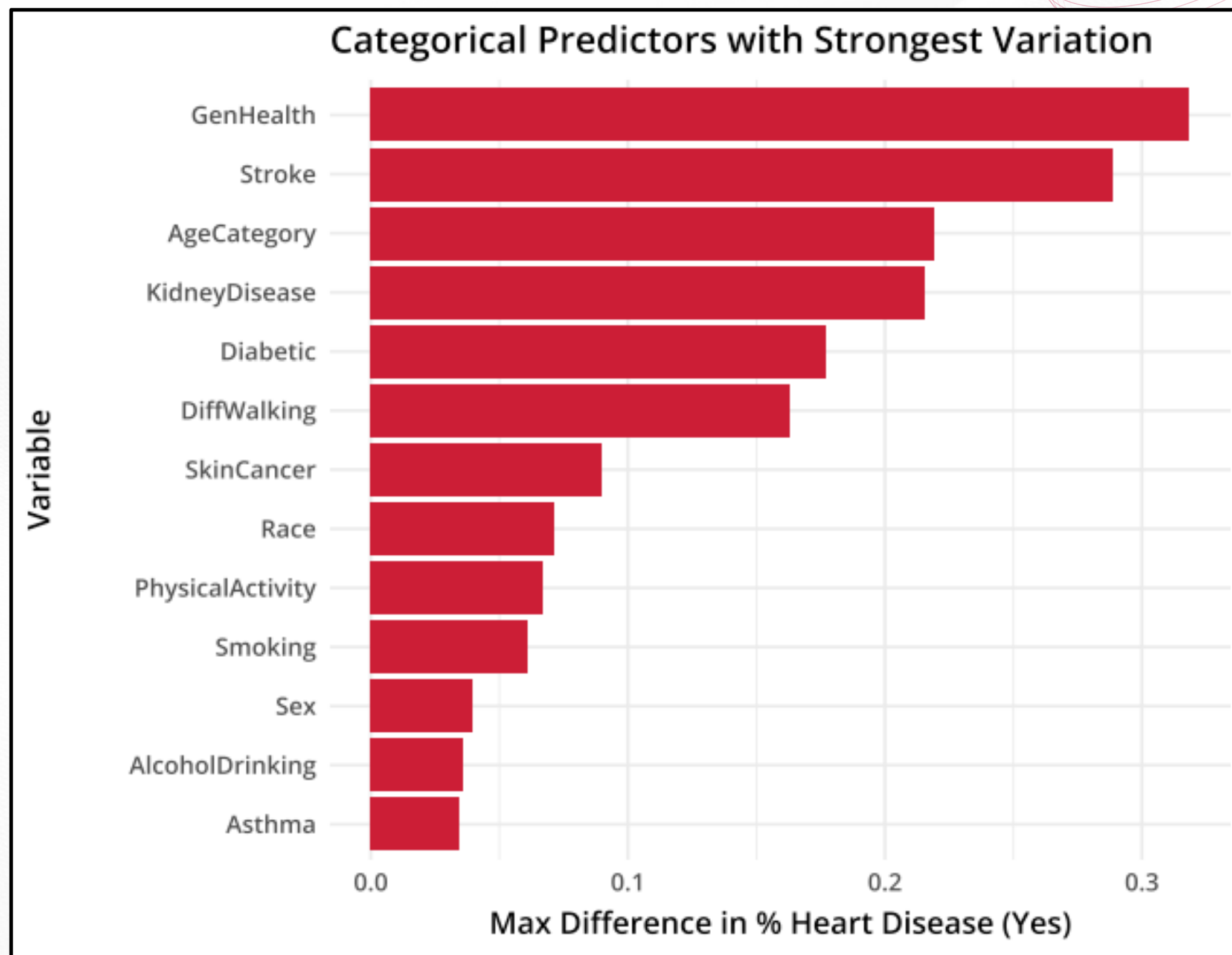
**ONLY NUMERIC
PREDICTORS ON
A 10% RANDOM
SAMPLE**

Scatterplot Matrix



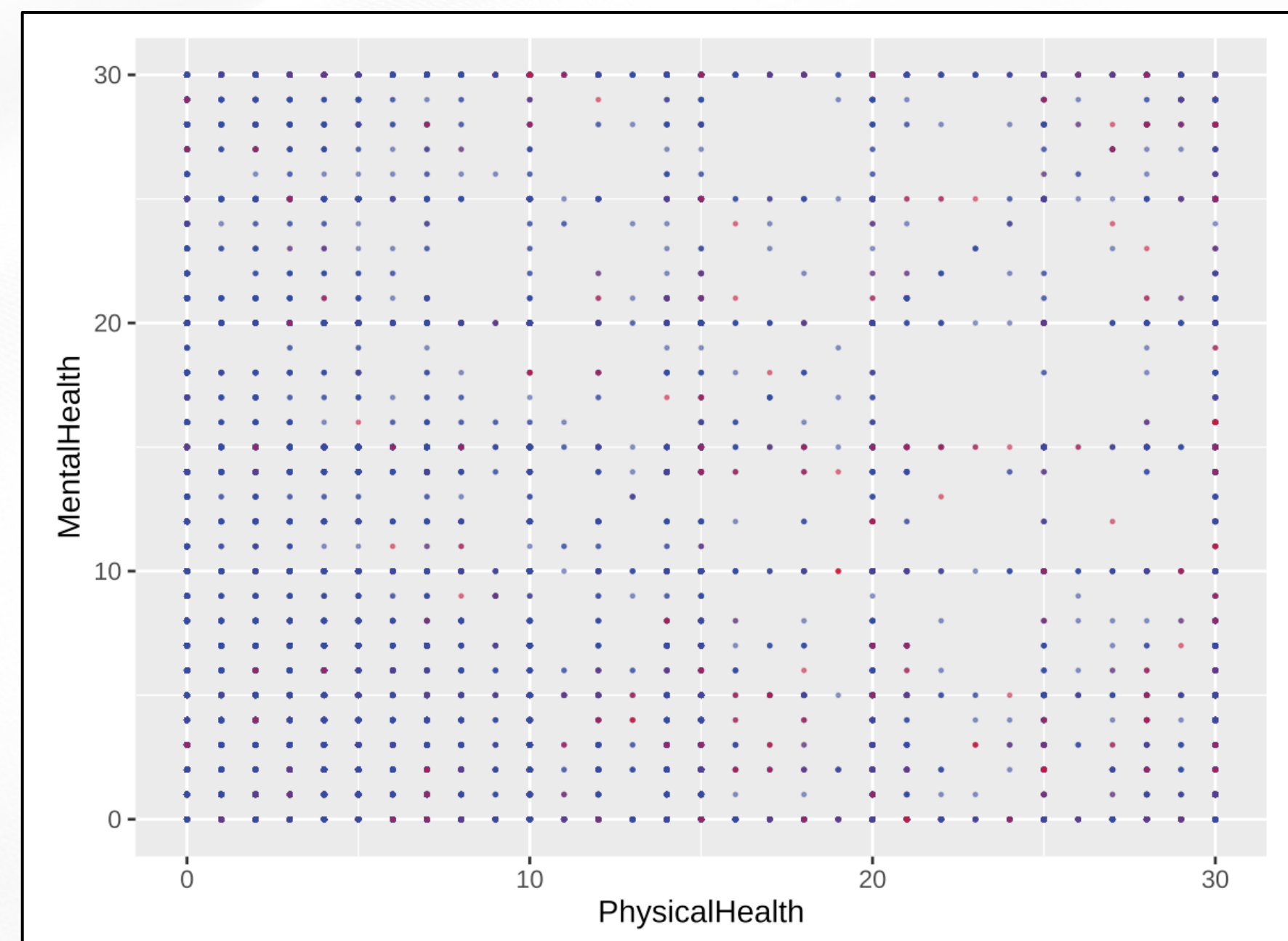
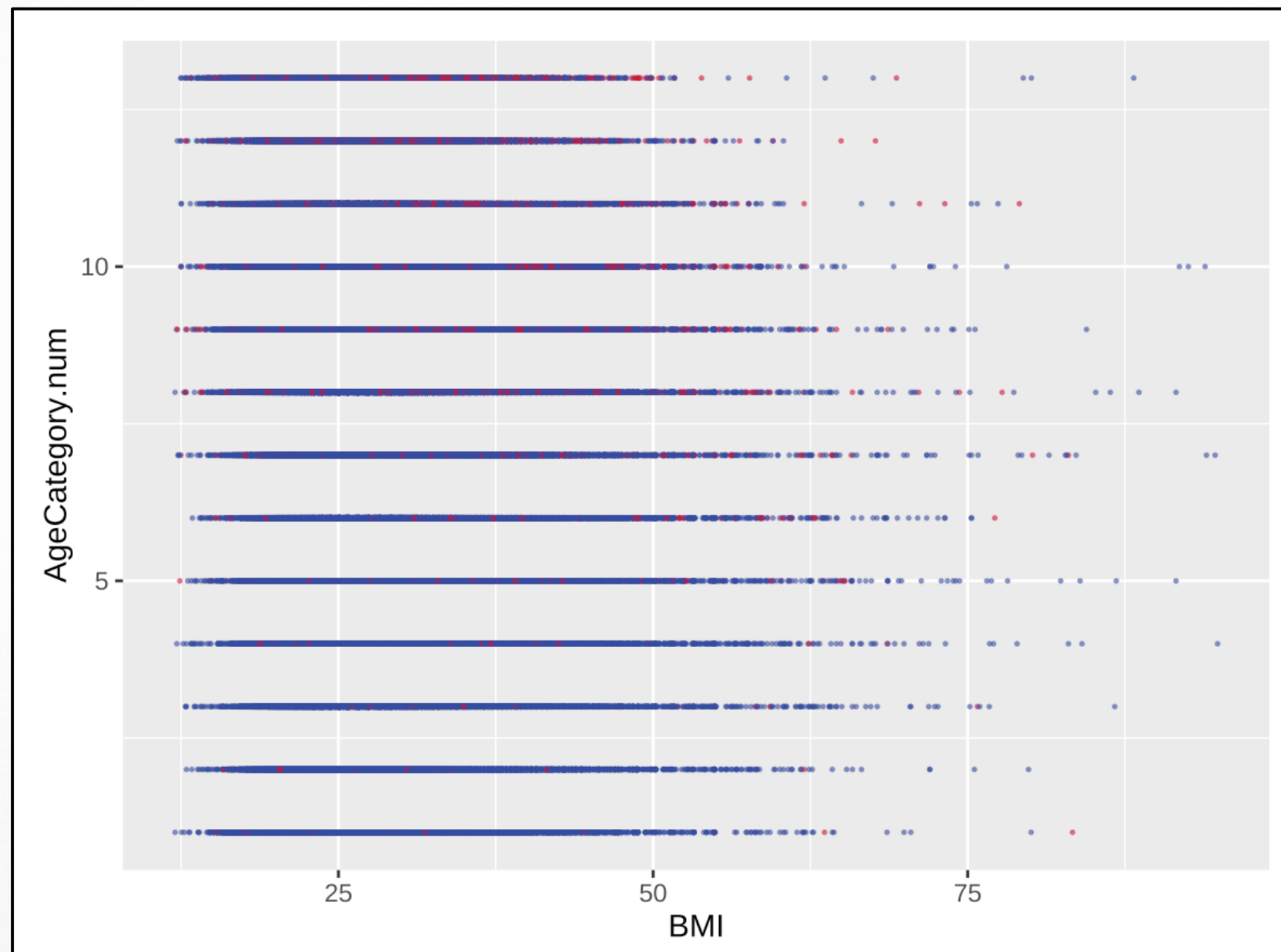
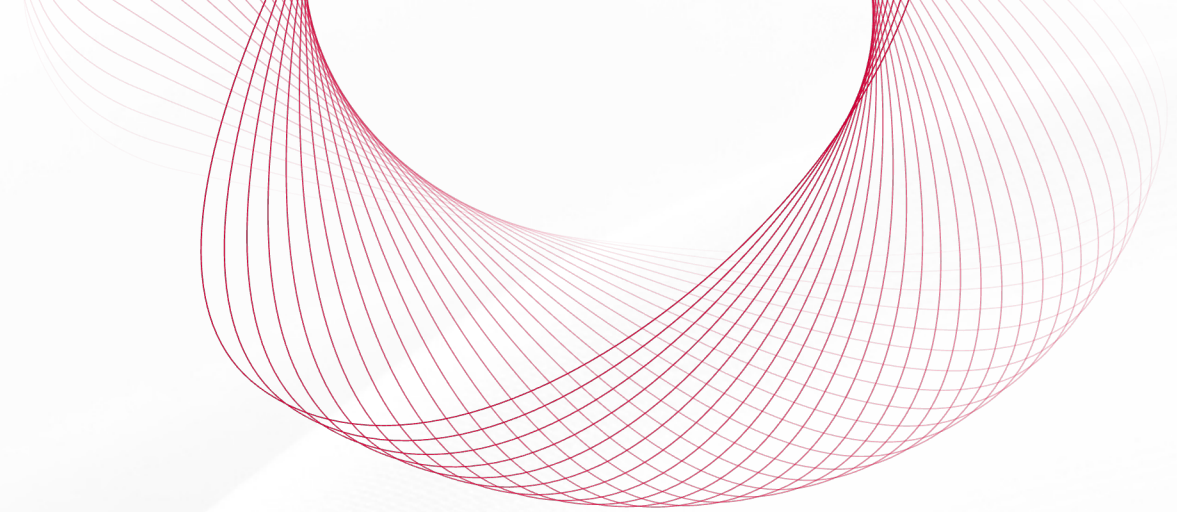


EDA Continued





Potential Interactions





Key Metrics for Heart Disease Classification

- **Sensitivity (True Positive Rate)**: Measures the proportion of actual heart disease cases correctly identified.
- **Specificity (True Negative Rate)**: Measures the proportion of non-cases correctly identified as not having heart disease.
- **PPV (Positive Predictive Value)**: Among those predicted to have heart disease, the proportion who actually do.
- **NPV (Negative Predictive Value)**: Among those predicted not to have heart disease, the proportion who actually don't.
- **Prevalence**: Class imbalance.
- **AUROC (AUC of ROC Curve)**: Measures how well the model distinguishes between classes across all thresholds.

Sensitivity	$TP / (TP + FN)$
Specificity	$TN / (TN + FP)$
PPV	$TP / (TP + FP)$
NPV	$TN / (TN + FN)$



Non-Additive Complex Logistic Regression Model

1 GLMNET FEATURE SELECTION

Resampling Results Across Tuning Parameters:

Alpha	Lambda	LogLoss
0.1	0.00011361	0.2257652

Polynomial Terms:

- BMI (up to degree 2)
- PhysicalHealth (up to degree 2)
- MentalHealth (up to degree 2)
- SleepTime², SleepTime³

Interaction Terms:

Polynomials of above numeric variables interacted with:

- GenHealth
- AgeCategory

Additional interactions:

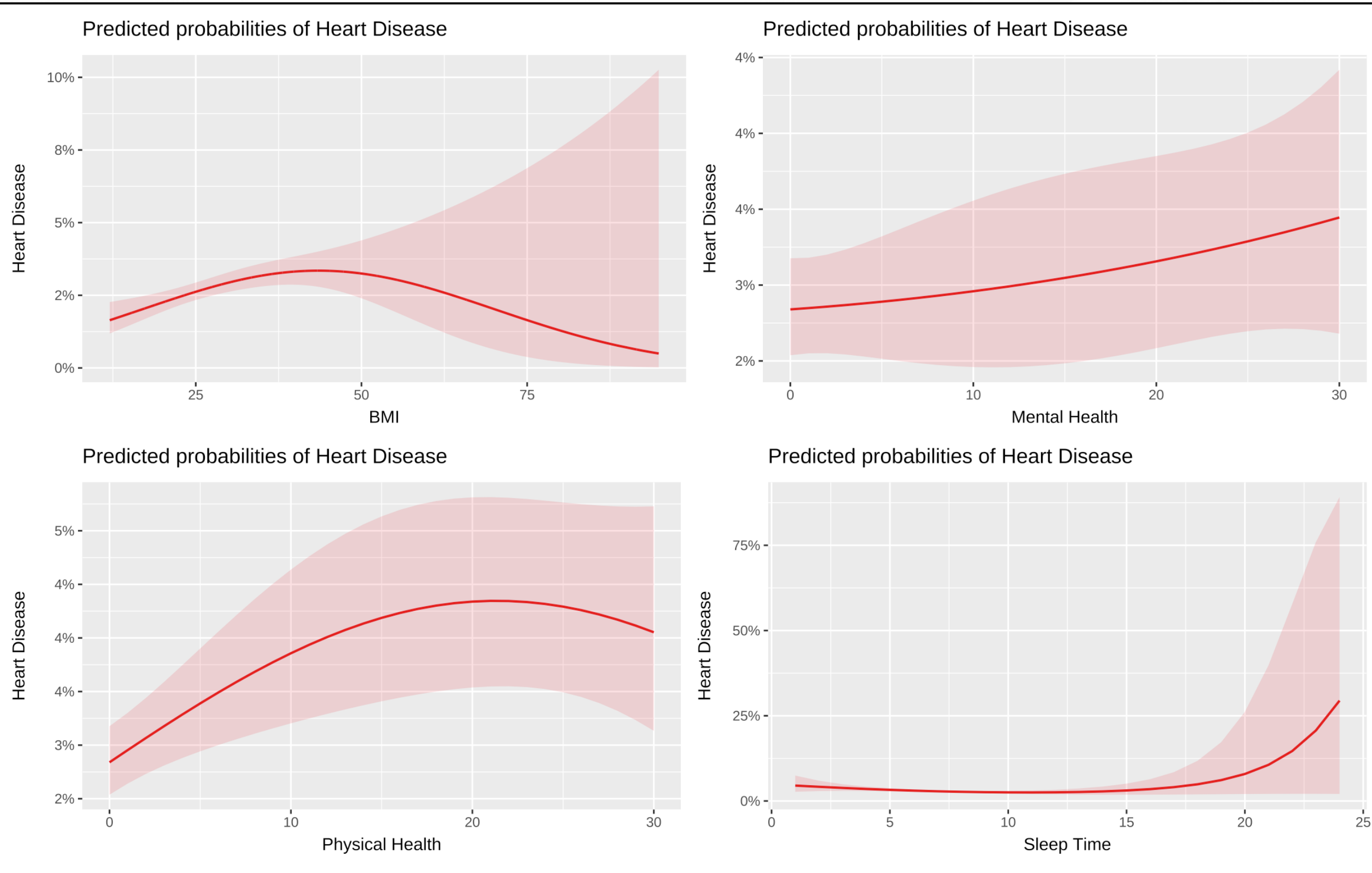
- GenHealth × AgeCategory
- GenHealth × Sex
- AgeCategory × Sex

Additive Categorical Predictors:

- AgeCategory
- GenHealth
- Smoking
- AlcoholDrinking
- Stroke
- DiffWalking
- Sex
- Race
- Diabetic
- PhysicalActivity
- Asthma
- KidneyDisease
- SkinCancer

AIC: 115,451.8

Effect Plots





kNN

1 Variable Selection

- Physical Health
- BMI
- Mental Health
- Sleep Time

2 Confusion Matrix

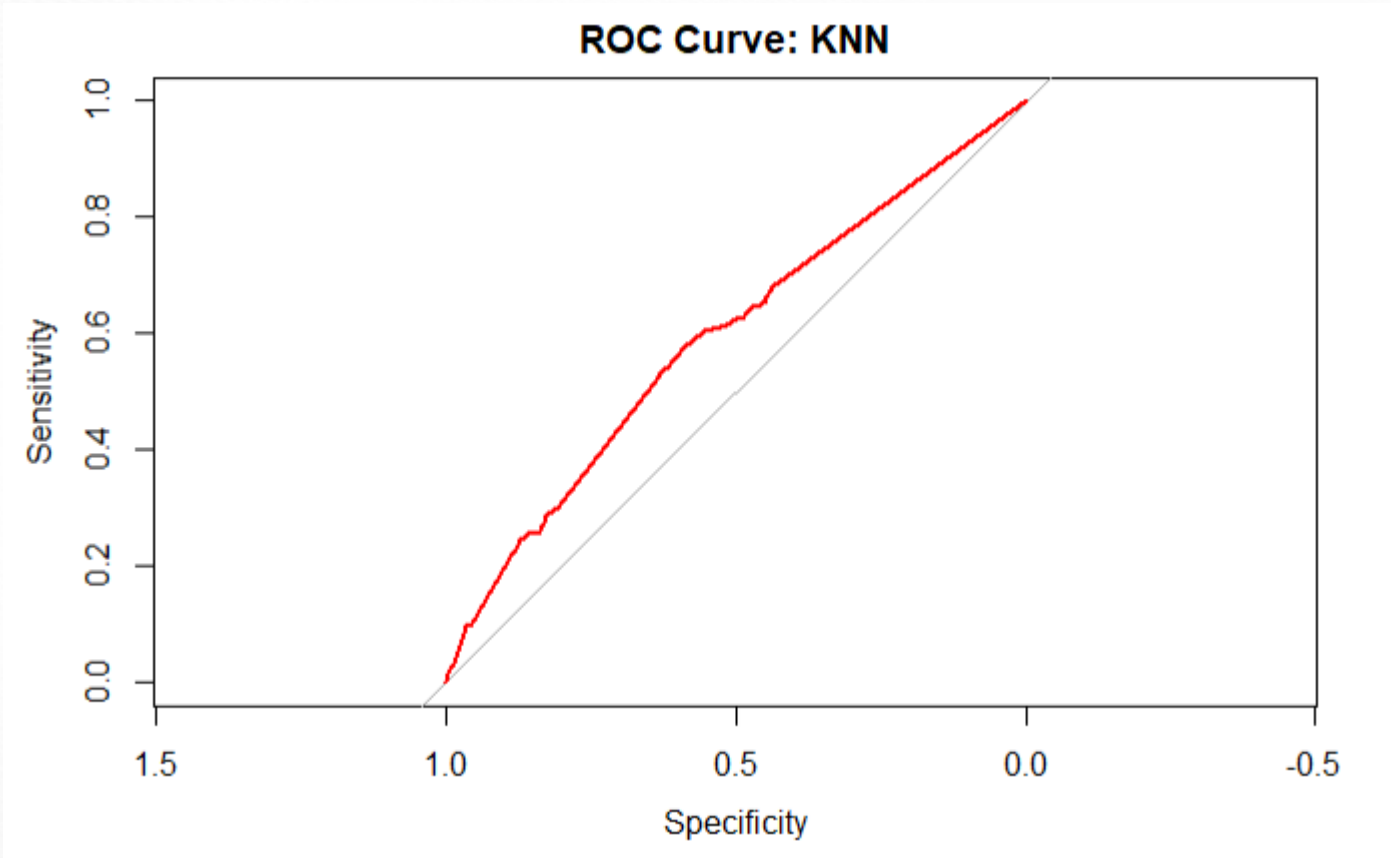
		Predicted	
		No	Yes
Actual	No	23086	1833
	Yes	355	309

Actual

3 AUC

.5917

4 ROC Curve



Linear and Quadratic Discriminant Analysis

LDA/QDA Assumptions

LDA Assumptions

- $X_+ \sim MVN(\mu_+, \Sigma)$
- $X_- \sim MVN(\mu_-, \Sigma)$
- Same variance-covariance matrix for the response classes
- Linear decision boundaries

QDA Assumptions

- $X_+ \sim MVN(\mu_+, \Sigma)$
- $X_- \sim MVN(\mu_-, \Sigma)$
- Variance-covariance matrices differ for each response class
- Conic decision boundaries - Quadratic (curved) separation

• Overfitting dampens with sample size

- Optimal method if assumptions are met
- Predictor sets can have different mean vectors for each response
- Neither model can include categorical predictors.
- Impacted by outliers





LDA

1

Variable Selection

Physical Health
BMI
Mental Health
Sleep Time

2

Confusion Matrix

		Predicted	
		No	Yes
Actual	No	22213	1794
	Yes	1228	348

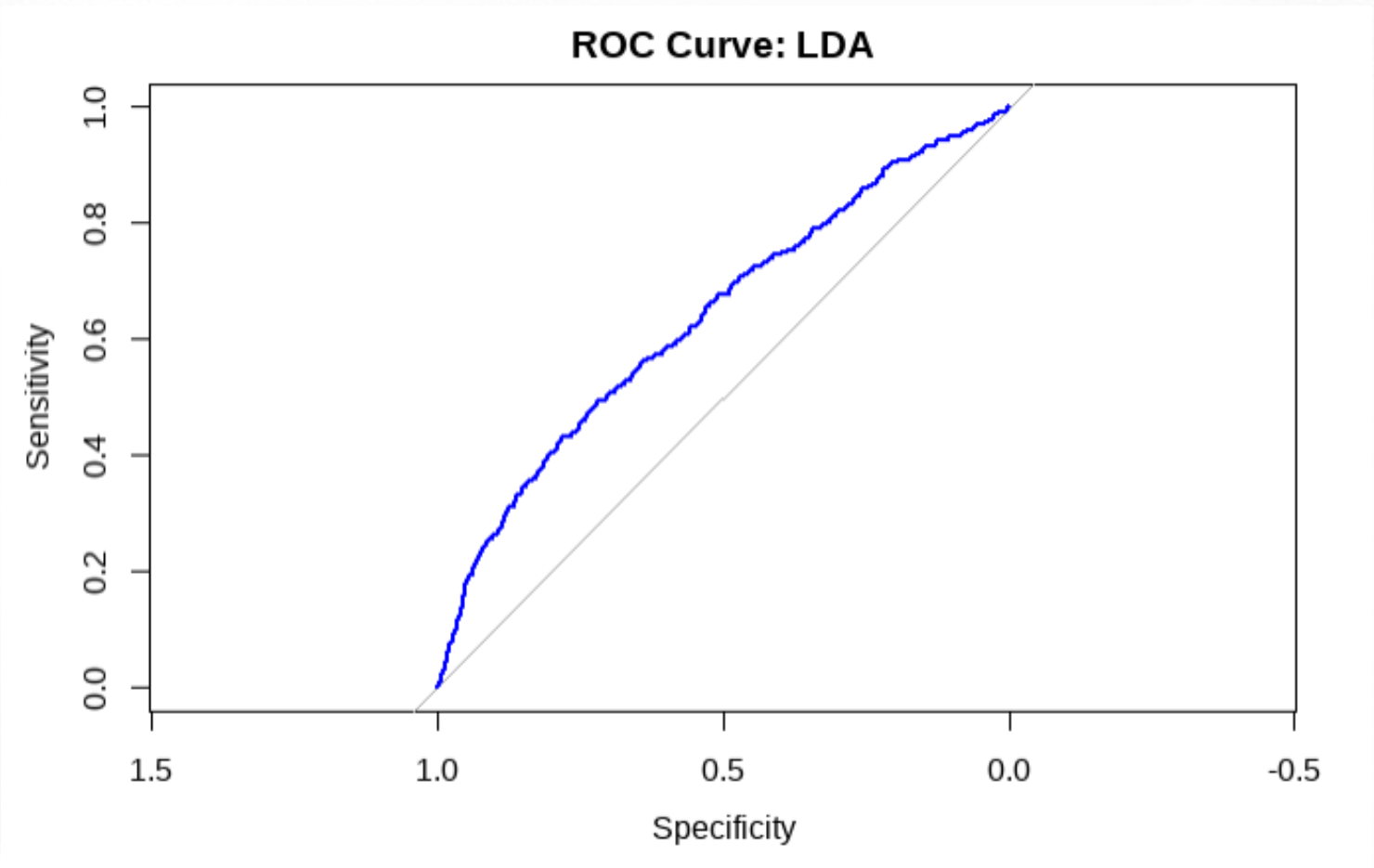
3

AUC

.6366

4

ROC Curve





QDA

1

Variable Selection

Physical Health
BMI
Mental Health
Sleep Time

2

Confusion Matrix

		Predicted	
		No	Yes
Actual	No	21746	1702
	Yes	1695	440

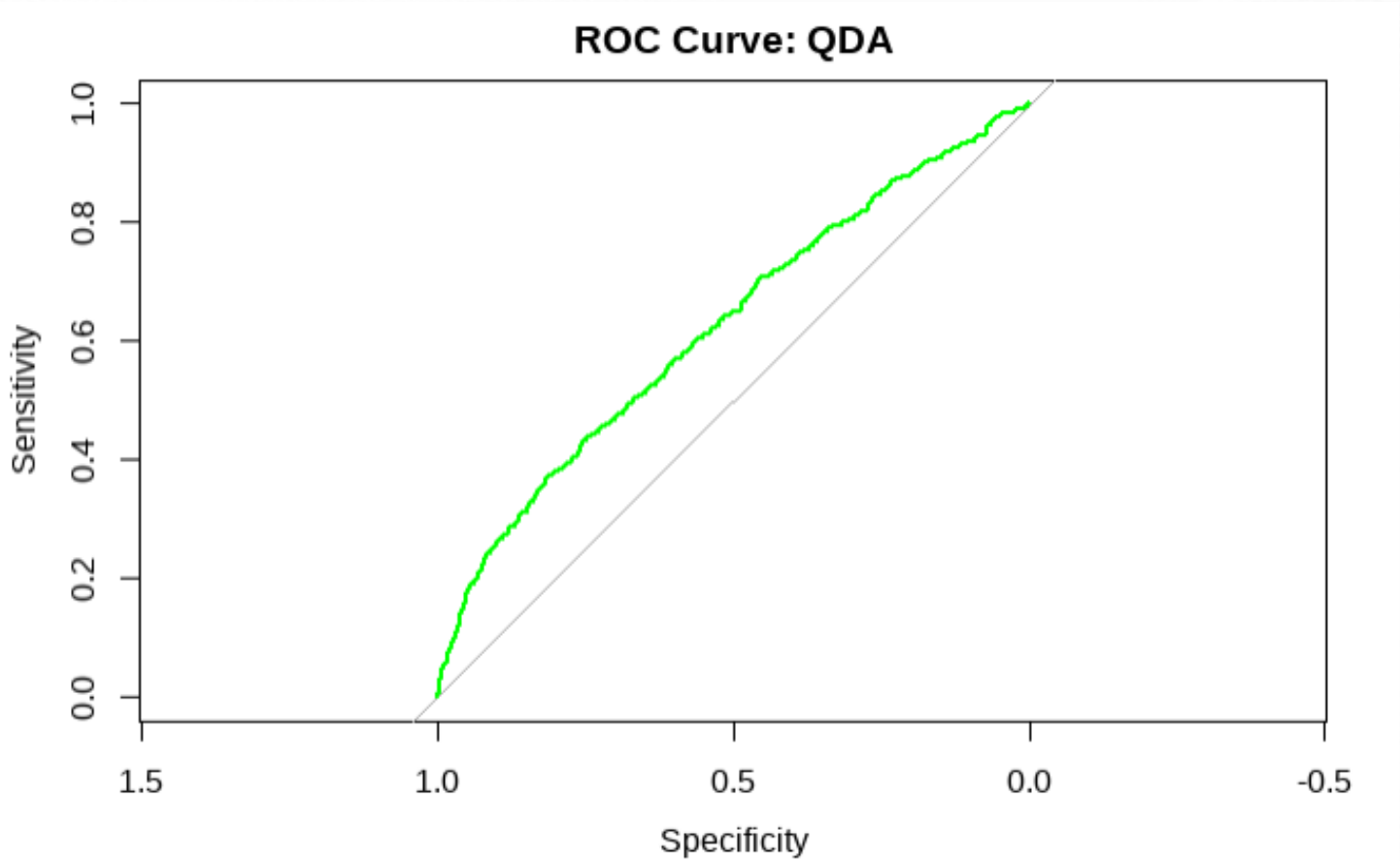
3

AUC

.6223

4

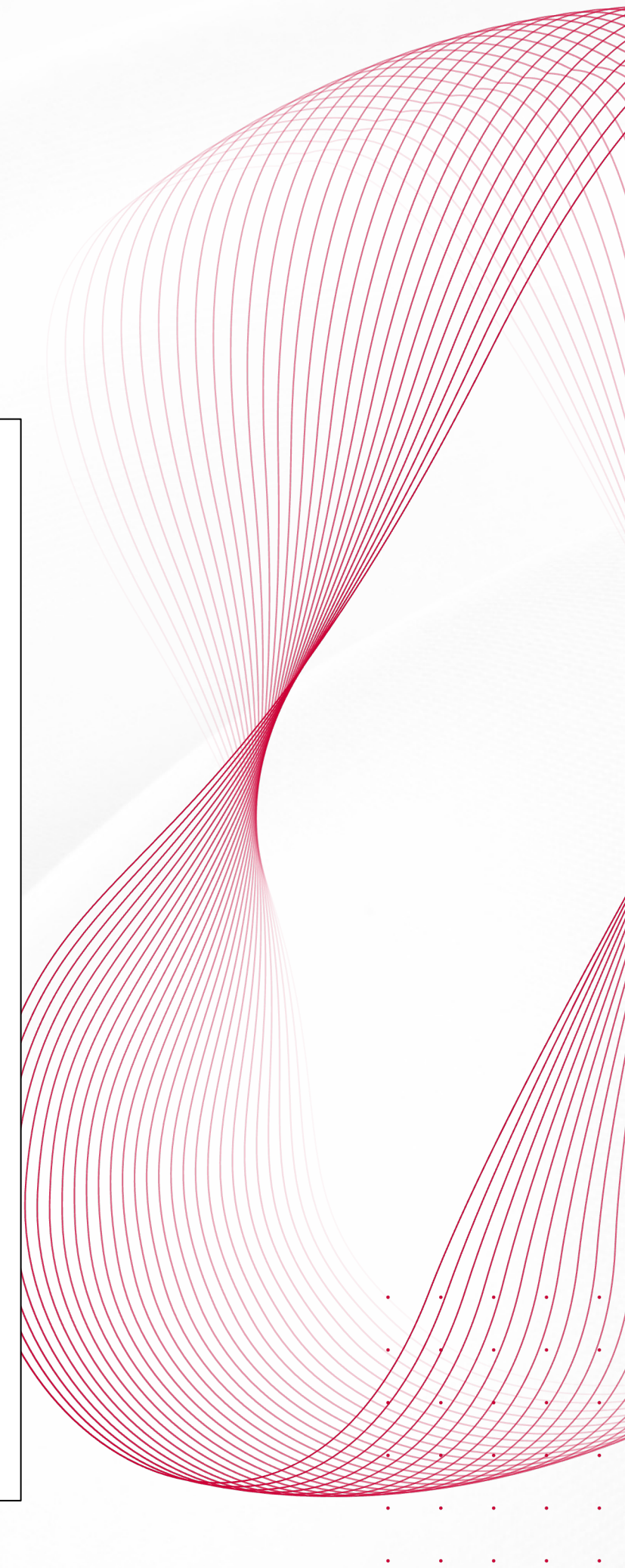
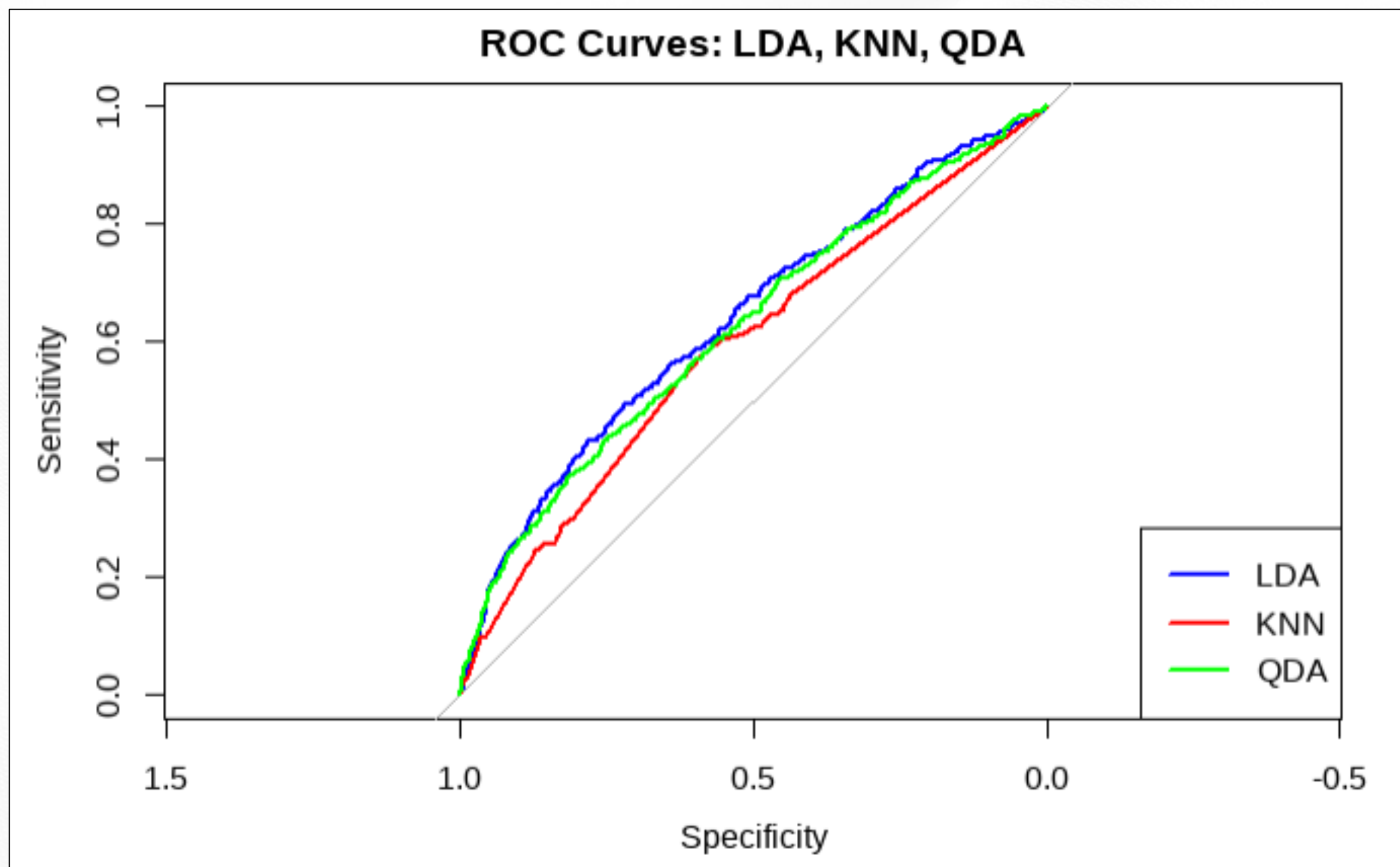
ROC Curve





KNN/LDA/QDA ROC Curves

Model Comparison



Model Comparison

Metric	Additive Logistic Regression Model	Complex - Non-Additive Model	QDA	LDA	KNN
Sensitivity	.2975	.3043	.2054	.1625	.1443
Specificity	.9607	.9603	.9277	.9476	.9849
Prevalence	.0856	.0856	.0837	.0837	.0837
PPV (Positive Predictive Value)	.4148	.4177	.2061	.2208	.4654
NPV (Negative Predictive Value)	.9359	.9365	.9274	.9253	.9264
AUROC (Area Under ROC Curve)	.8419	.8448	.6223	.6366	.5856

Conclusions

OBJECTIVE 1

Key Relevant Predictors

- Physical Activity was not statistically significant.
- All predictors were found to be practically significant.

OBJECTIVE 2

Best Overall Model

- Logistic Regression
- LDA had the highest AUC.
- Interpretability vs. Predictive Accuracy

LOGISTICAL CONCERNS

Future Recommendations

The data set are self-reported metrics, possibly biased by the perspective of the subject and their view on their own health. For future data, collection methods can benefit from targeting raw metrics such as cholesterol, blood pressure, and others.



Scope

This is an **observational study**, so although we found statistically significant evidence of an **association** between certain predictors and heart disease, we cannot infer causation. The BRFSS uses random-digit dialing (RDD) to recruit participants, but because this is a convenience sample and not every member of the population has an equal chance of being selected for our final analytic dataset, it is not a random sample. Therefore, these findings **cannot be generalized** to the broader U.S. adult population and only to the individuals in the sample or similar to those in this sample.





Thank You For Your Attention!

We work closely with you to ensure our solutions align with your vision and objectives.

Contact Information

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